1. What is HTTP?

HTTP (Hypertext Transfer Protocol) is a set of rules that allows a client, such as a browser, and a web server to communicate.

How it works

1. The user enters a website URL.
2. The browser sends an HTTP request to the server.
3. The server processes the request.
4. The server sends an HTTP response.
5. The browser displays the result.

2. What is HTTPS?

HTTPS (Hypertext Transfer Protocol Secure) is HTTP secured using TLS (Transport Layer Security).

It works like HTTP, but protects communication by encrypting data in transit, checking its integrity, and helping verify the server's identity.

Why do we use HTTPS when HTTP already exists?

HTTP allows a browser and a server to communicate with each other. However, HTTP does not encrypt data in transit. If someone intercepts the network traffic, they may be able to read or modify the data.

Real-life example

Imagine you are booking an appointment on a hospital website.

If the website uses HTTP:

- You enter your name and phone number.
- Your data travels without encryption.
- An attacker who intercepts the traffic may be able to read your personal information.

If the website uses HTTPS:

- Your information is encrypted using TLS while travelling between your browser and the server.
- An attacker who intercepts the traffic generally cannot read the original information.
- Unauthorized modifications to the data in transit can be detected.

Why was HTTP created?

HTTP was designed to provide rules for communication between browsers and web servers. However, secure communication also requires encryption, integrity protection, and authentication. HTTPS provides these protections through TLS.

Difference between HTTP and HTTPS

HTTP	HTTPS
Does not encrypt data in transit	Encrypts data in transit using TLS
Intercepted data may be readable	Intercepted data is generally unreadable
Default port: 80	Default port: 443
Not suitable for sensitive data without additional protection	Recommended for modern websites and APIs

Interview answer

HTTP is used for communication between a browser and a server, but it does not encrypt data in transit. HTTPS uses TLS to encrypt communication, protect data from tampering, and help verify the server's identity. That is why we use HTTPS instead of HTTP for modern websites, especially when handling sensitive information.